

# Digilians

## Univariate Analysis

# Univariate Analysis

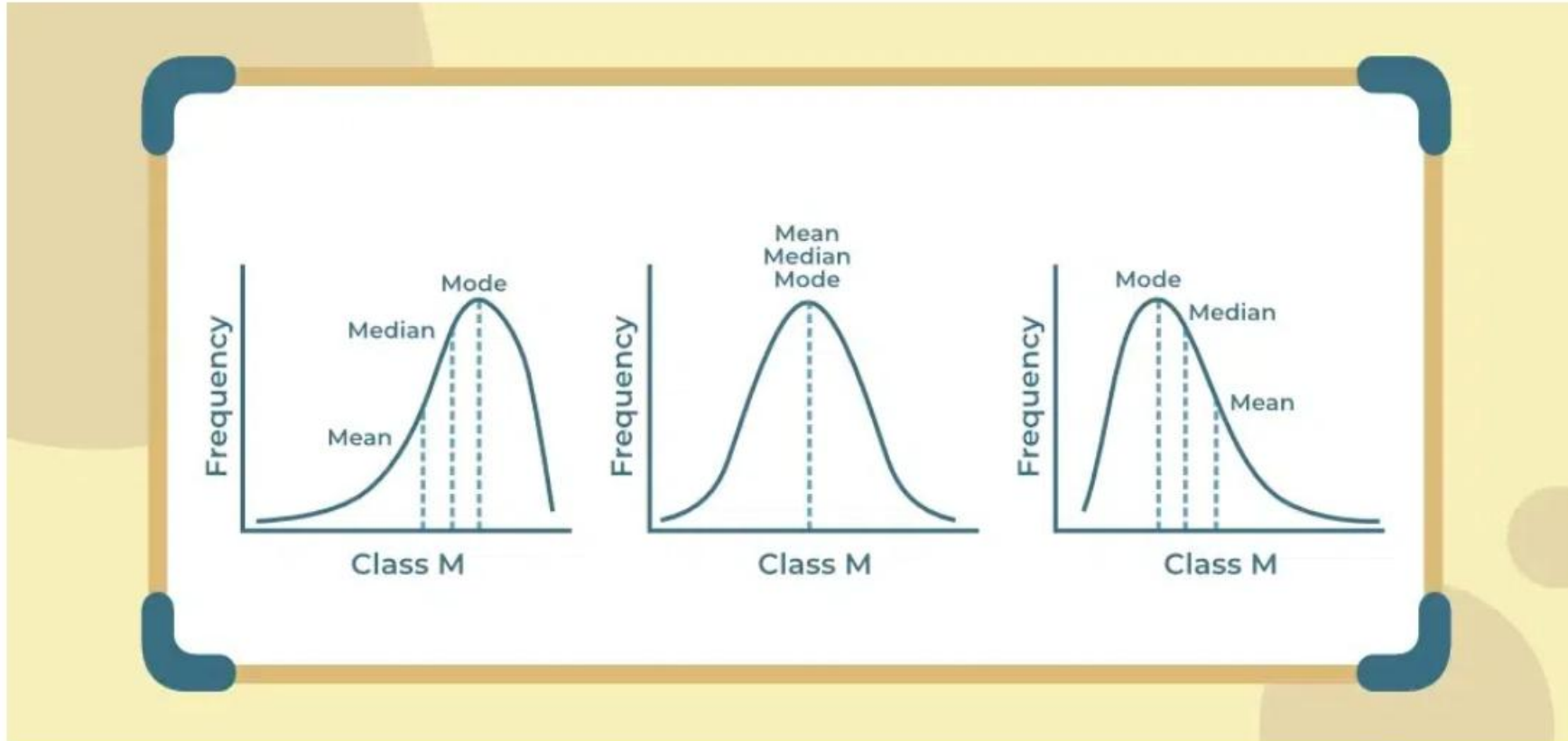
- **Measures of Central Tendency**
- **Measures of spread, IQR**
- **Skewness & Kurtosis**
- **Visualization: Histograms, Boxplots, Q-Q plots**

# Measures of Central Tendency in Statistics

- Central tendencies in statistics are numerical values that represent the middle or typical value of a dataset.
- Also known as averages, they provide a summary of the entire data, making it easier to understand the overall pattern or behavior.
- These values are useful because they capture the essence of large datasets in a single, representative number.
- *The representative value of a data set, generally the central value or the most occurring value that gives a general idea of the whole data set is called Measure of Central Tendency.*

## Central Tendency

The three most commonly used measures of central tendency are **mean**, **median**, and **mode**.



## Mean

Mean in general terms is used for the arithmetic mean of the data, but other than the arithmetic mean there are geometric mean and harmonic mean as well that are calculated using different formulas.

# Mean

$$\boxed{2} + \boxed{3} + \boxed{5} + \boxed{6} = \boxed{\frac{16}{4}} = \boxed{4}$$

The Mean Number is 4

## Mean for Ungrouped Data

Arithmetic mean ( $\bar{x}$ ) is defined as the sum of the individual observations ( $x_i$ ) divided by the total number of observations  $N$ . In other words, the mean is given by the sum of all observations divided by the total number of observations.

$$\bar{x} = \frac{\sum x_i}{N}$$

OR

*Mean = Sum of all Observations ÷ Total number of Observations*

**Example:** If there are 5 observations, which are 27, 11, 17, 19, and 21, then the mean ( $\bar{x}$ ) is given by

$$\bar{x} = (27 + 11 + 17 + 19 + 21) \div 5$$

$$\Rightarrow \bar{x} = 95 \div 5$$

$$\Rightarrow \bar{x} = 19$$

## Mean for Grouped Data

Mean ( $\bar{x}$ ) is defined for the grouped data as the sum of the product of observations ( $x_i$ ) and their corresponding frequencies ( $f_i$ ) divided by the sum of all the frequencies ( $f_i$ ).

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

**Example:** If the values ( $x_i$ ) of the observations and their frequencies ( $f_i$ ) are given as follows:

|       |   |    |    |    |    |
|-------|---|----|----|----|----|
| $x_i$ | 4 | 6  | 15 | 10 | 9  |
| $f_i$ | 5 | 10 | 8  | 7  | 10 |

then Arithmetic mean ( $\bar{x}$ ) of the above distribution is given by

$$\begin{aligned}\bar{x} &= (4 \times 5 + 6 \times 10 + 15 \times 8 + 10 \times 7 + 9 \times 10) \div (5 + 10 + 8 + 7 + 10) \\ \Rightarrow \bar{x} &= (20 + 60 + 120 + 70 + 90) \div 40 \\ \Rightarrow \bar{x} &= 360 \div 40 \\ \Rightarrow \bar{x} &= 9\end{aligned}$$



# Types of Mean

- Mean can be classified into three different class groups, which are
- Arithmetic Mean
- Geometric Mean
- Harmonic Mean

## Arithmetic Mean:

The Arithmetic Mean is the most common type of average. It is obtained by adding all the observations and then dividing by the total number of observations. It gives a simple average value representing the entire data set.

The formula for the Arithmetic Mean is given by

$$\bar{x} = \frac{\sum x_i}{N}$$

Where,

- $x_1, x_2, x_3, \dots, x_n$  are the observations, and
- $N$  is the number of observations.

## Geometric Mean:

The Geometric Mean is used when the data values are multiplicative or vary exponentially, such as in growth rates, interest rates, or ratios. It is the  $n$ th root of the product of all observations.

The formula for Geometric Mean is given by

$$\text{G.M.} = \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n}$$

Where,

- $x_1, x_2, x_3, \dots, x_n$  are the observations, and
- $n$  is the number of observations.

## Harmonic Mean:

The Harmonic Mean is appropriate when the data consists of rates or ratios, such as speed, time, or efficiency.

It is calculated as the reciprocal of the arithmetic mean of reciprocals of the data values.

The formula for Harmonic Mean is given by

$$H. M. = \frac{n}{1/x_1 + 1/x_2 + \dots + 1/x_n}$$

OR

$$H. M. = \frac{n}{\sum (1/x_i)}$$

Where,

- $x_1, x_2, \dots, x_n$  are the observations, and
- $n$  is the number of observations.

## Properties of Mean (Arithmetic)

There are various properties of Arithmetic Mean, some of which are as follows:

- The algebraic sum of deviations from the arithmetic mean is zero, i.e.,  $\sum (x_i - \bar{x}) = 0$ .
- If  $\bar{x}$  is the arithmetic mean of observations, and  $a$  is added to each of the observations, then the new arithmetic mean is given by  $\bar{x}' = \bar{x} + a$
- If  $\bar{x}$  is the arithmetic mean of observations, and  $a$  is subtracted from each of the observations, then the new arithmetic mean is given by  $\bar{x}' = \bar{x} - a$
- If  $\bar{x}$  is the arithmetic mean of observations, and  $a$  is multiplied by each of the observations, then the new arithmetic mean is given by  $\bar{x}' = \bar{x} \times a$
- If  $\bar{x}$  is the arithmetic mean of observations, and each of the observations is divided by  $a$ , then the new arithmetic mean is given by  $\bar{x}' = \bar{x} \div a$

Case 1: When N is Odd

*Median = Value of observation at  $[(n + 1) \div 2]^{\text{th}}$  Position*

When N is odd the median is calculated as shown in the image below.

## Median



Median of Data Set when N is odd

*Median when N is Odd*

### Case 2: When N is Even

*Median = Arithmetic mean of Values of observations at  $(n \div 2)^{th}$  and  $[(n \div 2) + 1]^{th}$  Position*

When N is even the median is calculated as shown in the image below.

## Median



When N is even Median =  $(7 + 9)/2 = 8$

**Example 1:** If the observations are 25, 36, 31, 23, 22, 26, 38, 28, 20, 32, then the Median is given by

*Arranging the data in ascending order: 20, 22, 23, 25, **26, 28**, 31, 32, 36, 38*

*$N = 10$  which is even then*

*Median = Arithmetic mean of values at  $(10 \div 2)$ th and  $[(10 \div 2) + 1]$ th position*

*$\Rightarrow \text{Median} = (\text{Value at 5th position} + \text{Value at 6th position}) \div 2$*

*$\Rightarrow \text{Median} = (26 + 28) \div 2$*

*$\Rightarrow \text{Median} = 27$*



**Example 2:** If the observations are 25, 36, 31, 23, 22, 26, 38, 28, 20, then the Median is given by

*Arranging the data in ascending order: 20, 22, 23, 25, **26**, 28, 31, 36, 38*

*$N = 9$  which is odd then*

*Median = Value at  $[(9 + 1) \div 2]^{\text{th}}$  position*

*$\Rightarrow$  Median = Value at  $5^{\text{th}}$  position*

*$\Rightarrow$  Median = 26*

## Median of Grouped Data

Median of Grouped Data is given as follows:

$$\text{Median} = l + \frac{N/2 - c_f}{f} \times h$$

Where,

- $l$  is the lower limit of the median class,
- $n$  is the total number of observations,
- $c_f$  is the cumulative frequency of the preceding class,
- $f$  is the frequency of each class, and
- $h$  is the class size.

**Example:** Calculate the median for the following data.

| Class     | 10 - 20 | 20 - 30 | 30 - 40 | 40 - 50 | 50 - 60 |
|-----------|---------|---------|---------|---------|---------|
| Frequency | 5       | 10      | 12      | 8       | 5       |

**Solution:**

*Create the following table for the given data.*

| <i>Class</i> | <i>Frequency</i> | <i>Cumulative Frequency</i> |
|--------------|------------------|-----------------------------|
| 10 - 20      | 5                | 5                           |
| 20 - 30      | 10               | 15                          |
| 30 - 40      | 12               | 27                          |
| 40 - 50      | 8                | 35                          |
| 50 - 60      | 5                | 40                          |

As  $n = 40$  and  $n/2 = 20$ ,  
Thus, 30 - 40 is the median class.  
 $l = 30$ ,  $c_f = 15$ ,  $f = 12$ , and  $h = 10$

Putting the values in the formula  $\text{Median} = l + \frac{N/2 - c_f}{f} \times h$

$$\text{Median} = 30 + (20 - 15)/12 \times 10$$

$$\Rightarrow \text{Median} = 30 + (5/12) \times 10$$

$$\Rightarrow \text{Median} = 30 + 4.17$$

$$\Rightarrow \text{Median} = 34.17$$

**So, the median value for this data set is 34.17**

## Mode

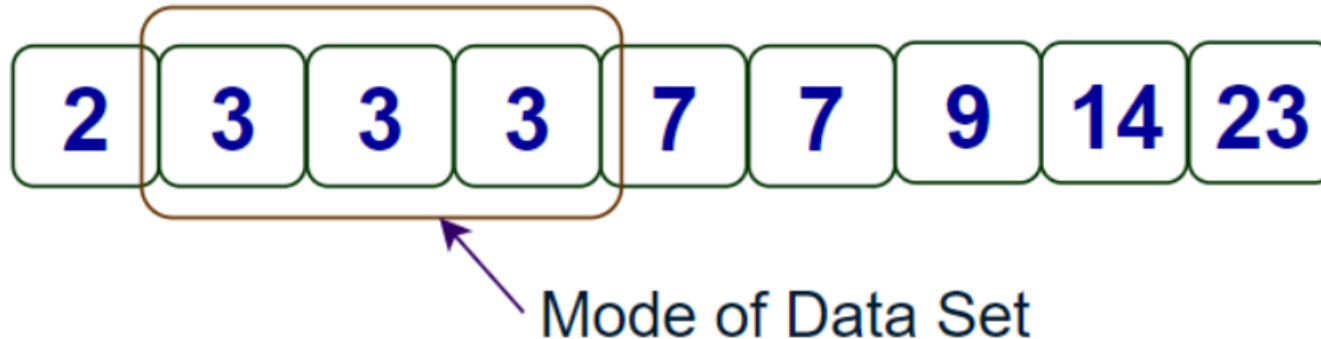
Mode is the value of that observation which has a maximum frequency corresponding to it. In other, that observation of the data occurs the maximum number of times in a dataset.

### Mode of Ungrouped Data

Mode of Ungrouped Data can be simply calculated by observing the observation with the highest frequency. Let's see an example of the calculation of the mode of ungrouped data.

The mode of the data set is the highest frequency term in the data set, as shown in the image added below.

## Mode



**Example:** Find the mode of observations 5, 3, 4, 3, 7, 3, 5, 4, 3.

**Solution:**

*Create a table with each observation with its frequency as follows:*

|       |   |   |   |   |
|-------|---|---|---|---|
| $x_i$ | 5 | 3 | 4 | 7 |
| $f_i$ | 2 | 4 | 2 | 1 |

*Since 3 has occurred a maximum number of times i.e. 4 times in the given data;*

***Hence, Mode of the given ungrouped data is 3.***

## Mode of Grouped Data

Formula to find the mode of the grouped data is:

$$\text{Mode} = l + \left[ \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

Where,

- $l$  is the lower class limit of modal class,
- $h$  is the class size,
- $f_1$  is the frequency of modal class,
- $f_0$  is the frequency of class that proceeds the modal class, and
- $f_2$  is the frequency of class that succeeds the modal class.

**Example:** Find the mode of the dataset which is given as follows.

| Class Interval | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
|----------------|-------|-------|-------|-------|-------|
| Frequency      | 5     | 8     | 12    | 16    | 10    |

**Solution:**

*As the class interval with the highest frequency is 40-50, which has a frequency of 16. Thus, 40-50 is the modal class.*

*Thus,  $l = 40$  ,  $h = 10$  ,  $f_1 = 16$  ,  $f_0 = 12$  ,  $f_2 = 10$*

*Plugging in the values in formula  $\text{Mode} = l + \left[ \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$ , we get*

$$\text{Mode} = 40 + (16 - 12)/(2 \times 16 - 12 - 10) \times 10$$

$$\Rightarrow \text{Mode} = 40 + (4/10) \times 10$$

$$\Rightarrow \text{Mode} = 40 + 4$$

$$\Rightarrow \text{Mode} = 44$$

**Therefore, the mode for this set of data is 44.**



## Measures of Central Tendency: Formulas Table

The formulas for calculating mean, median, and mode vary slightly depending on whether the data is ungrouped or grouped. The table below summarizes the key formulas for each measure of central tendency:

| Measure | Ungrouped Data Formula                                                                                                                                                                                              | Grouped Data Formula                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Mean    | $\bar{x} = \frac{\sum x_i}{N}$                                                                                                                                                                                      | $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$                                      |
| Median  | Median = Value of observation at $[(n + 1) \div 2]^{\text{th}}$ Position (Odd)<br>Median = Arithmetic mean of Values of observations at $(n \div 2)^{\text{th}}$ and $[(n \div 2) + 1]^{\text{th}}$ Position (Even) | $\text{Median} = l + \frac{N/2 - c_f}{f} \times h$                             |
| Mode    | Most frequent value                                                                                                                                                                                                 | $\text{Mode} = l + \left[ \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$ |

## Empirical Relation Between Measures of Central Tendency

The three central tendencies are related to each other by the empirical formula, which is given as follows:

$$2 \times \text{Mean} + \text{Mode} = 3 \times \text{Median}$$

This formula is used to calculate one of the central tendencies when two other central tendencies are given.

# Measures of Spread

- Measures of spread (also called measures of variability or dispersion) describe how spread out or scattered a set of data values is. While measures of central tendency (like mean or median) tell you the center, measures of spread tell you how much the data varies around that center.

# Range

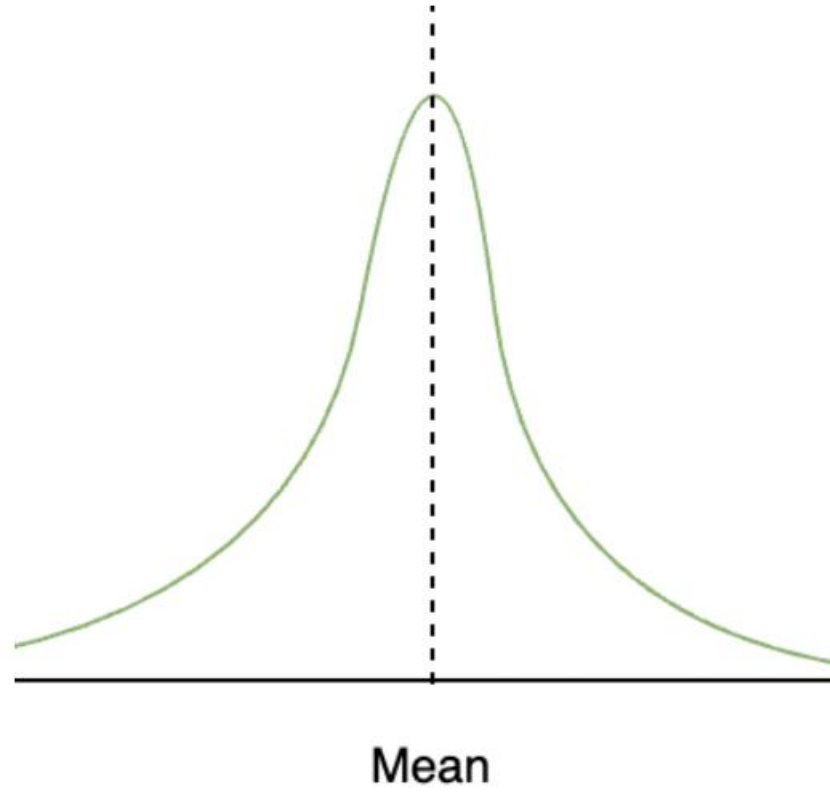
- The range of the data is given as the difference between the maximum and the minimum values of the observations in the data. For example, let's say we have data on the number of customers walking in the store in a week.
- 10, 14, 8, 10, 15, 4, 7
- Minimum value in data = 4
- Maximum Value in the data = 15

- Range = Maximum Value in the data - Minimum value in the data
- = 15 – 4
- = 11
- Now we can say that the range of the data is 8. This gives us an idea about the spread of the data but doesn't tell how the data is distributed.

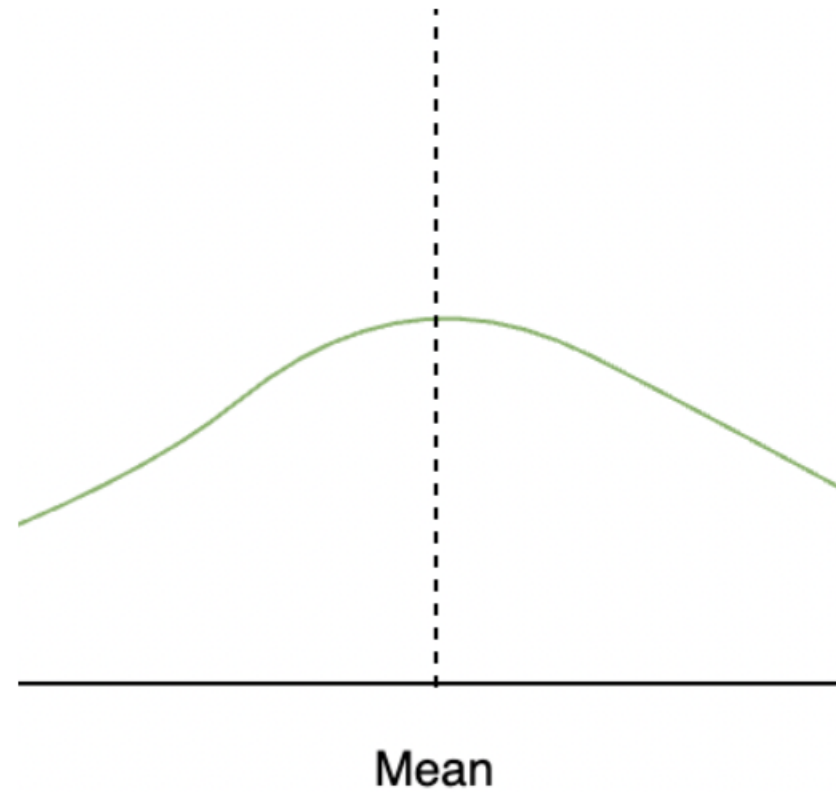
# Variance

- The variance of the data is given by measuring the distance of the observed values from the mean of the distribution.
- Here we are not concerned with the sign of the distance of the point, we are more interested in the magnitude.
- So, we take squares of the distance from the mean. Let's say we have  $x_1, x_2, x_3, \dots, x_n$  as  $n$  observations and  $\bar{x}$  be the mean.

$$(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2 \dots (x_n - \bar{x})^2 = \sum_0^n (x_i - \bar{x})^2$$



If this sum zero, then each term has to be zero which means that there is no scattering in the data. If it is small, then it means that the data is concentrated at the mean and vice versa for large values of the variance.





But this measure is still dependent on the number of observations in the data. That is if there are lots of observations this value will become large. So, we take the mean of the data,

$$\text{Variance} = \frac{\sum_0^n (x_i - \bar{x})^2}{n}$$

$$\sigma^2 = \frac{\sum_0^n (x_i - \bar{x})^2}{n}$$

## Standard Deviation

In the calculation of variance, notice that the units of the variance and the unit of the observations are not the same. So, to remove this problem, we define standard deviation. It is denoted as  $\sigma$

$$\sigma = \sqrt{\frac{\sum_0^n (x_i - \bar{x})^2}{n}}$$

Let's see how to calculate these measures in some problems,

Question 1: Find out the range of the following data:

|    |   |     |   |   |
|----|---|-----|---|---|
| -4 | 5 | -10 | 6 | 9 |
|----|---|-----|---|---|

Solution:

*For calculating the range of a data, we need to find out the maximum and the minimum of the data:*

$$\text{Max} = 9$$

$$\text{Min} = -10$$

$$\text{Range} = \text{Max} - \text{Min}$$

$$= 9 - (-10)$$

$$= 19$$

Question 2: Find out the mean and the median of the same data:

|    |   |     |   |   |
|----|---|-----|---|---|
| -4 | 5 | -10 | 6 | 9 |
|----|---|-----|---|---|

Solution:

$$\text{Mean of the data: } \frac{(-4+5-10+6+9)}{5}$$

$$= \frac{6}{5}$$

$$= 1.2$$

Median is called the middle element of the data.

Median = -10.

Question 3: Let's say we have the following data,

|    |    |   |    |   |   |   |   |    |   |
|----|----|---|----|---|---|---|---|----|---|
| -4 | -2 | 0 | -2 | 6 | 4 | 6 | 0 | -6 | 4 |
|----|----|---|----|---|---|---|---|----|---|

Calculate the range, variance, and standard deviation of the data.

### Range

*We need to find out the minimum and the maximum values of the data distribution.*

*Min Value = -6*

*Max Value = +6*

*Range = Max Value - Min Value*

$$= 6 - (-6)$$

$$= 12$$

### Variance

*To find the variance, we first need to find the mean,*

$$\text{Mean} = \frac{(-4 + -2 + 0 + -2 + 6 + 4 + 6 + 0 - 6 + 4)}{10}$$

$$= \frac{6}{10}$$

$$= 0.6$$

*We know the formula for Variance,*

$$\begin{aligned}\text{Variance} &= \frac{\sum_0^n (x_i - \bar{x})^2}{n} \\ &= \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n} \\ &= \frac{(-4 - 0.6)^2 + (-2 - 0.6)^2 + (0 - 0.6)^2 + \dots}{10} \\ &= 17.82\end{aligned}$$

### *Standard Deviation*

$$\sigma = \sqrt{\frac{\sum_0^n (x_i - \bar{x})^2}{n}}$$

$$\Rightarrow \sigma = \sqrt{17.82}$$

$$\Rightarrow \sigma = 4.22$$

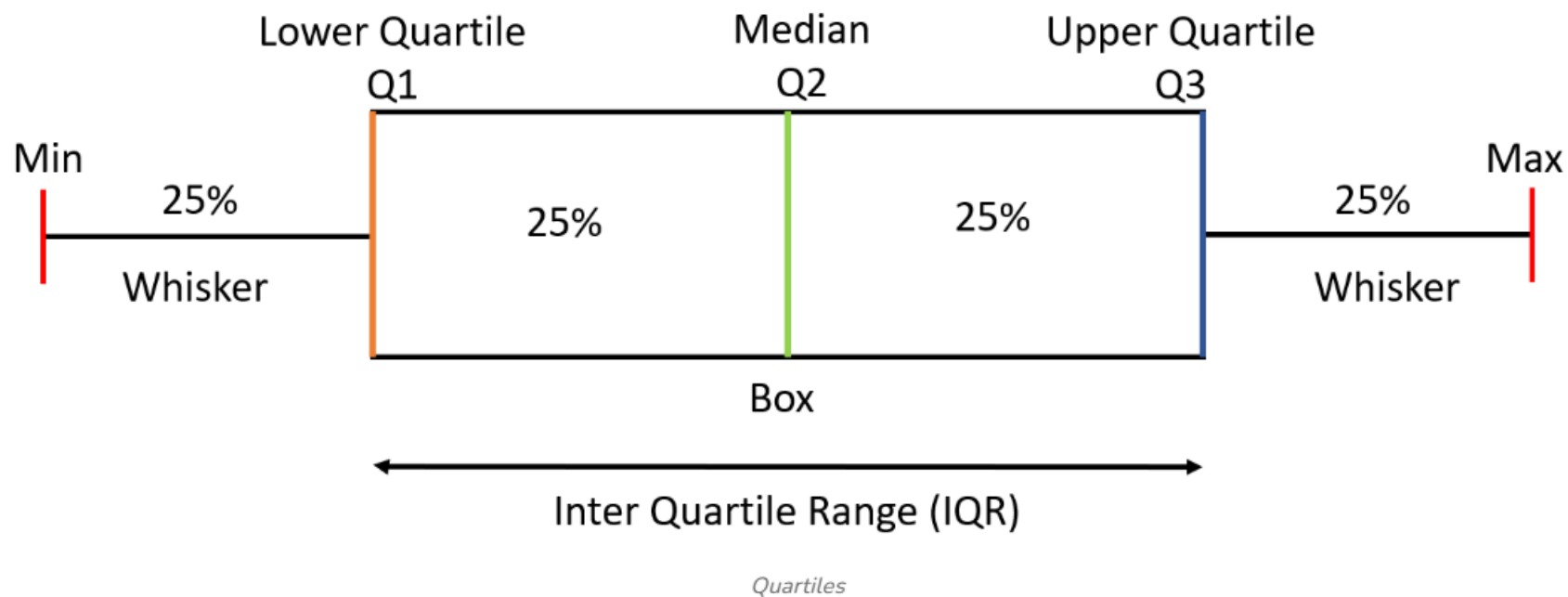
# Interquartile Range and Quartile Deviation

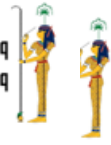
- In statistical analysis, understanding the spread or variability of a dataset is crucial for gaining insights into its distribution and characteristics.
- Two common measures used for quantifying this variability are the interquartile range (IQR) and quartile deviation.



# Quartiles

- Quartiles are a kind of quantile that divides the number of data points into four parts, or quarters.
- The **first quartile (Q1)** , is defined as the middle number between the smallest number and the median of the data set,
- The **second quartile (Q2)** is the median of the given data set.
- The **third quartile (Q3)** is the middle number between the median and the largest value of the data set.



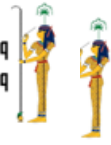


- **Algorithm to find Quartiles**
- Here's a step-by-step algorithm to find quartiles:
- Sort the dataset in ascending order.
- Calculate the total number of entries in the dataset.

- If the number of entries is even:
  - Calculate the median ( $Q_2$ ) by taking the average of the two middle values.
  - Divide the dataset into two halves: the first half containing the smallest  $n$  entries and the second half containing the largest  $n$  entries, where  $n = \text{total number of entries} / 2$ .
  - Calculate  $Q_1$  as the median of the first half.
  - Calculate  $Q_3$  as the median of the second half.

- If the number of entries is odd:
  - Calculate the median ( $Q_2$ ) as the middle value.
  - Divide the dataset into two halves: the first half containing the smallest  $n$  entries and the second half containing the largest  $n$  entries, where  $n = (\text{total number of entries} - 1) / 2$ .
  - Calculate  $Q_1$  as the median of the first half.
  - Calculate  $Q_3$  as the median of the second half.
- The calculated values of  $Q_1$ ,  $Q_2$ , and  $Q_3$  represent the first quartile, median (second quartile), and third quartile respectively.

- **Range:**
- It is the difference between the largest value and the smallest value in the given data set.
- **Interquartile Range**
- The interquartile range (IQR) is indeed defined as the difference between the third quartile (Q3) and the first quartile (Q1). It's often used as a measure of statistical dispersion, specifically focusing on the middle 50% of the data. This range effectively covers the center of the distribution and contains 50% of the observations, making it useful for understanding the variability within a dataset while being less sensitive to outliers compared to the range.



- Mathematically, it's represented as:
- $IQR = Q_3 - Q_1$
- Where ,  $Q_3$  is the third quartile and  $Q_1$  is the first quartile.

- **Uses of IQR**

- The interquartile range has a breakdown point of 25% due to which it is often preferred over the total range.
- The IQR is used to build box plots, simple graphical representations of a probability distribution.
- The IQR can also be used to identify the outliers in the given data set.
- The IQR gives the central tendency of the data.



- **Interpretation of IQR**

- The data set has a higher value of interquartile range (IQR) has more variability.
- The data set having a lower value of interquartile range (IQR) is preferable.
- Suppose, if we have two data sets and their interquartile ranges are  $IR1$  and  $IR2$ , and if  $IR1 > IR2$  then the data in  $IR1$  is said to have more variability than the data in  $IR2$  and data in  $IR2$  is preferable.

- **Quartile Deviation**

- The quartile deviation is a measure of statistical dispersion or spread within a dataset. It's defined as half of the difference between the third quartile (Q3) and the first quartile (Q1). Mathematically, it's represented as:

- Quartile Deviation = 
$$\frac{Q3 - Q1}{2}$$

- **Uses of Quartile Deviation**

- Quartile deviation quantifies spread within a dataset, computed as half the difference between the third and first quartiles.
- It provides a robust measure of variability less sensitive to outliers compared to other measures like the range or standard deviation.
- Used in descriptive statistics to complement measures of central tendency.
- Helps assess skewness and identify potential outliers in distributions.
- Facilitates comparison of variability between different datasets or subsets of data.

## • Interpretation of Quartile Deviation

- Quartile deviation represents the average spread or variability within the middle 50% of a dataset.
- It indicates how data points are distributed around the median, providing insights into the dispersion of values.
- A larger quartile deviation suggests greater variability among the central 50% of data points.
- Quartile deviation is less affected by extreme values or outliers compared to other measures of spread, making it robust in skewed distributions.
- It aids in comparing the consistency or dispersion of data between different datasets or subsets.

# How to Calculate Skewness and Kurtosis

- **Skewness** is a statistical term and it is a way to estimate or measure the shape of a distribution.
- It is an important statistical methodology that is used to estimate the asymmetrical behavior rather than computing frequency distribution. Skewness can be two types:



- **Symmetrical:** A distribution can be called symmetric if it appears the same from the left and right from the center point.
- **Asymmetrical:** A distribution can be called asymmetric if it doesn't appear the same from the left and right from the center point.

- **Distribution on the basis of skewness value:**
- **Skewness = 0:** Then normally distributed.
- **Skewness > 0:** Then more weight in the left tail of the distribution.
- **Skewness < 0:** Then more weight in the right tail of the distribution.

# Kurtosis:

- It is also a statistical term and an important characteristic of frequency distribution. It determines whether a distribution is heavy-tailed in respect of the normal distribution. It provides information about the shape of a frequency distribution.
- kurtosis for normal distribution is equal to 3.
- For a distribution having kurtosis  $< 3$ : It is called platykurtic.
- For a distribution having kurtosis  $> 3$ , It is called leptokurtic and it signifies that it tries to produce more outliers rather than the normal distribution.

•

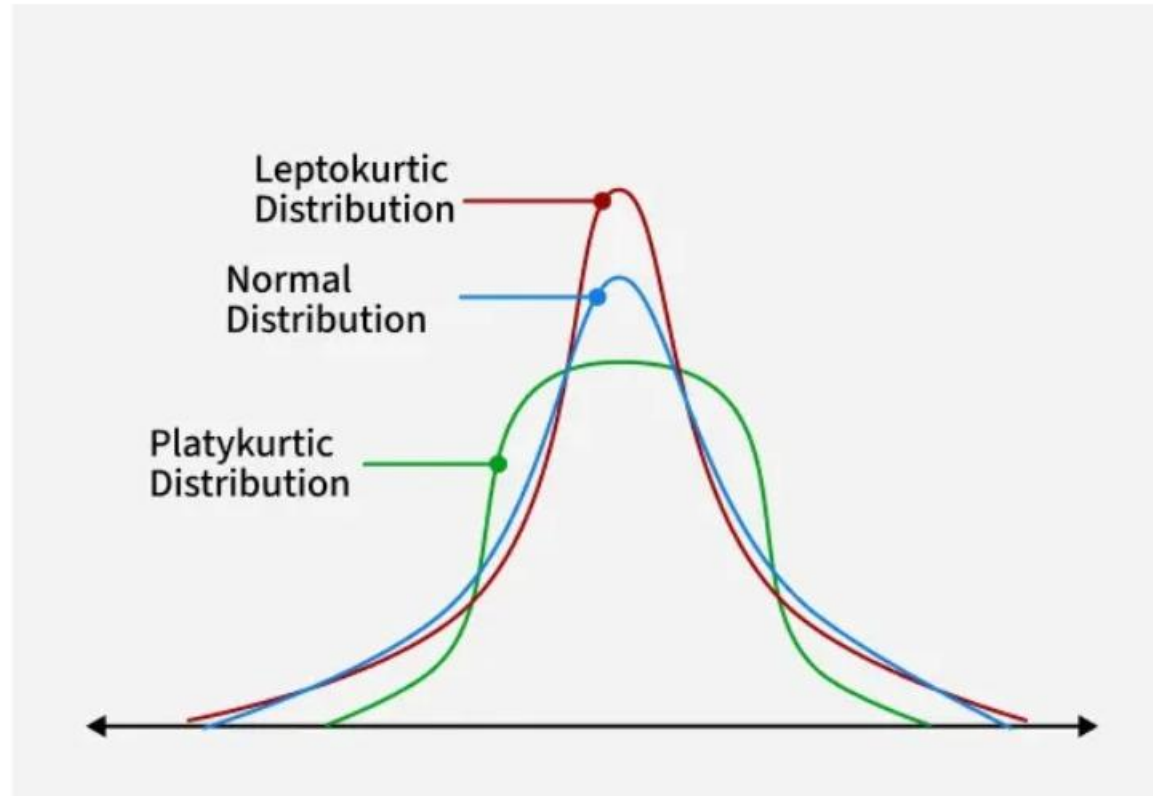


- **Positive Skewness:** It refers to a distribution where the right tail is longer or more stretched than the left. The majority of data points are concentrated on the lower-value side, while a few large values pull the mean to the right.
- **Negative skewness:** It refers to a distribution where the left tail is longer or more stretched than the right. Most data points lie on the higher-value side, while a few small values pull the mean to the left.
- **Zero Skewness:** It represents a perfectly symmetrical distribution, where the left and right sides mirror each other and extreme values occur equally on both sides.

- **Kurtosis**

- Kurtosis measures how sharp the peak of a distribution is and how heavy or light its tails are compared to a normal distribution. It shows how much of the data lies close to the center and how much appears in the extreme ends.
- Higher kurtosis implies more extreme outliers.
- Lower kurtosis implies fewer extreme values and smoother tails.
- Excess kurtosis ( $\text{kurtosis} - 3$ ) is commonly used for interpretation.

## Types of Kurtosis



- **Mesokurtic:** Mesokurtic distributions have a kurtosis value around 3, indicating a moderate peak and tails similar to a normal distribution. They do not show unusually heavy or light tails.
- **Leptokurtic:** A leptokurtic distribution has kurtosis greater than 3. It has a sharp, narrow peak and heavy tails, meaning the data clusters tightly near the mean but also produces more
- **Platykurtic:** A platykurtic distribution has kurtosis less than 3. It features a flatter, wider peak and light tails, indicating fewer extreme values and more evenly spread data.

## Skewness vs. Kurtosis

Let's compare skewness and kurtosis,

| Aspect                       | Skewness                                                   | Kurtosis                                                                |
|------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------|
| Core Purpose                 | Quantifies the asymmetry of a distribution                 | Quantifies the peakedness and tail weight of a distribution             |
| Primary Focus                | Direction and degree of horizontal asymmetry               | Vertical structure and extremity of tail behaviour                      |
| Positive Values              | Reflect a distribution with a longer or heavier right tail | Reflect a sharper peak with heavier tails and more extreme values       |
| Negative Values              | Reflect a distribution with a longer or heavier left tail  | Reflect a flatter peak with lighter tails and fewer extreme values      |
| Effect on Summary Statistics | Alters the relationship among mean, median and mode        | Influences tail probability, variance concentration and outlier impact  |
| Sensitivity                  | Sensitive to directional deviations and skewed clusters    | Highly sensitive to occurrence and magnitude of extreme observations    |
| Analytical Implication       | Used to identify directional bias and underlying imbalance | Used to assess risk, extremity and departure from normal tail behaviour |
| Role in Distribution Shape   | Describes shape horizontally                               | Describes shape vertically and in the tails                             |

## Skewness Formulas

### 1. Moment-Based (Sample)

This is the most common definition, using the third central moment. 

- **Formula:** 
$$\text{Skewness} = \frac{M_3}{M_2^{3/2}} = \frac{\sum (x_i - \bar{x})^3 / n}{s^3}$$
  - $\bar{x}$  is the sample mean.
  - $s$  is the sample standard deviation.
  - $n$  is the number of observations.
  - $M_3$  is the third central moment,  $M_2$  is the variance (second central moment).

### 2. Alternative (Sample)

A common, simpler approximation for samples. 

- **Formula:** 
$$\text{Skewness} = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

## Kurtosis Formulas

### 1. Moment-Based (Sample)

Measures the fourth central moment relative to the variance, with excess kurtosis subtracting 3 for comparison to normal. [🔗](#)

- **Formula (Excess Kurtosis):** 
$$\text{Kurtosis} = \frac{\sum (x_i - \bar{x})^4 / n}{s^4} - 3$$
  - A normal distribution has an excess kurtosis of 0.
  - Positive values (leptokurtic) mean heavier tails/sharper peak; negative (platykurtic) mean lighter tails/flatter peak.

### 2. Moment-Based (Population)

For a population, the formula is similar but without sample adjustments. [🔗](#)

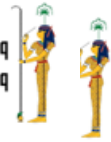
- **Formula:** 
$$\text{Kurtosis} = E \left[ \left( \frac{X - \mu}{\sigma} \right)^4 \right]$$

# Histogram | Meaning, Example, Types and Steps to Draw

- What is Histogram?
- A histogram is a graphical representation of the frequency distribution of continuous series using rectangles. The x-axis of the graph represents the class interval, and the y-axis shows the various frequencies corresponding to different class intervals. A histogram is a two-dimensional diagram in which the width of the rectangles shows the width of the class intervals, and the length of the rectangles depicts the corresponding frequency.



- There are no gaps between two consecutive rectangles based on the fact that histograms can be drawn when data are in the form of the frequency distribution of a **continuous series**. No histogram can be drawn for a data set in the form of discrete series, and this makes histograms different from bar graphs as they can be plotted for both discrete and continuous series. The major difference between a histogram and a bar graph is that the former is two-dimensional; i.e., both the width and length of the rectangles are used for comparison, whereas the latter is one-dimensional, which means only the length of the rectangles is used for comparison. A histogram is used to determine the value of **the Mode** of a data set in the form of a continuous series.



# Types of Histogram

- Histograms of Frequency Distribution are of two types:
- **Histogram of Equal Class Intervals**
- **Histogram of Unequal Class Intervals**

- **1. Histogram of Equal Class Intervals**
- When histograms are drawn based on the data with equal class intervals, they are known as Histograms of equal class intervals. The histogram of equal class intervals includes rectangles with equal width; however, the length of the rectangles is proportional to the frequency distribution of the class intervals.

### Example of Histogram of Equal Class Intervals:

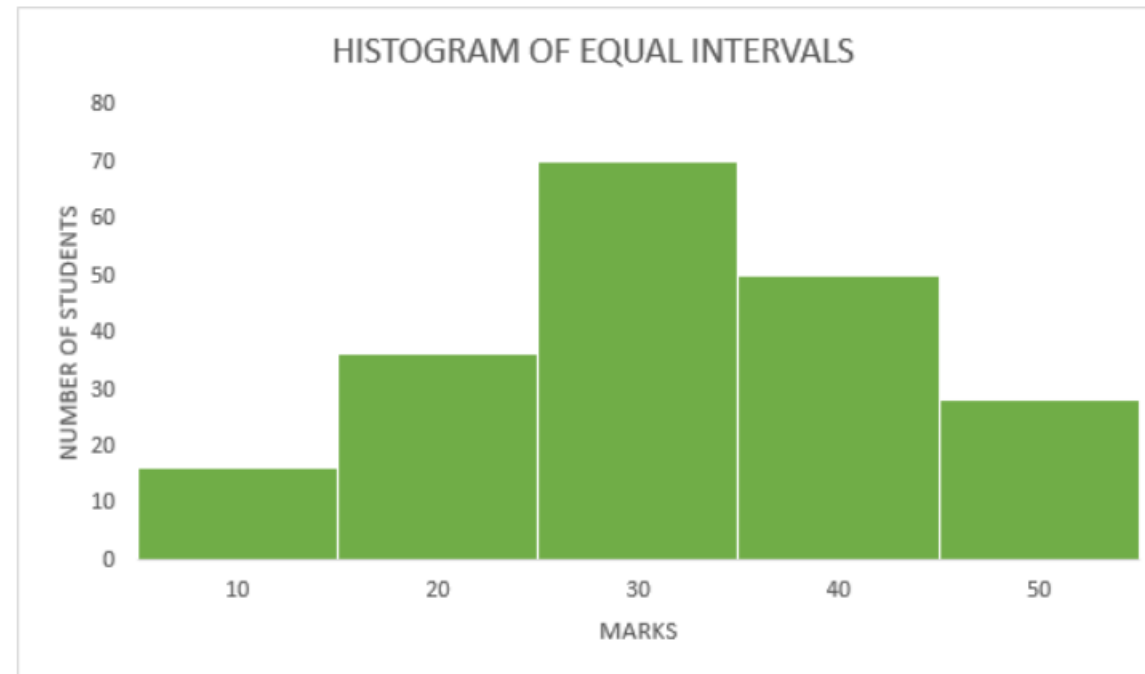
Present the following information in the form of a Histogram:

| Marks              | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
|--------------------|------|-------|-------|-------|-------|
| Number of Students | 16   | 36    | 70    | 50    | 28    |

## Solution

## Solution

- It is visible that the set of data given is of the equal class interval; i.e., the difference between the upper limit and the lower limit of each class interval is 10. So, drawing a Histogram is feasible.
- The X-axis represents the marks (class intervals), and Y-axis represents the number of students (frequency distribution).



## 2. Histogram of Unequal Class Intervals

When histograms are drawn based on the data with unequal class intervals, they are known as Histograms of unequal class intervals. Histogram of unequal class intervals includes rectangles of different width sizes. Therefore, before drawing a histogram in case of unequal class intervals, frequency distribution has to be adjusted.

### *Adjustment of frequencies of unequal class intervals:*

1. Determine the class of the smallest interval ( lowest class interval ).
2. Then, calculate the adjustment factor using the formula:

$$\text{Adjustment Factor for any Class} = \frac{\text{Class Interval of the Concerned Class}}{\text{Lowest Class Interval}}$$

3. Now, adjust the given frequencies using the adjustment factor:

$$\text{Frequency Density} = \frac{\text{Given Frequency}}{\text{Adjustment Factor}}$$

### Example of Histogram of Unequal Class Intervals:

Present the following information in the form of a Histogram:

| Wages | Number of Workers |
|-------|-------------------|
| 10-15 | 14                |
| 15-20 | 20                |
| 20-25 | 54                |
| 25-30 | 30                |
| 30-40 | 24                |
| 40-60 | 24                |
| 60-80 | 16                |

## Solution

1. It can be seen clearly that the given class interval is unequal. So, before plotting the histogram, frequencies have to be adjusted.
2. Determine the class of the smallest interval, i.e., 10-15. Thus, the lowest class interval in the given frequency distribution is 5.
3. Formulate the Adjusted Table as shown below:

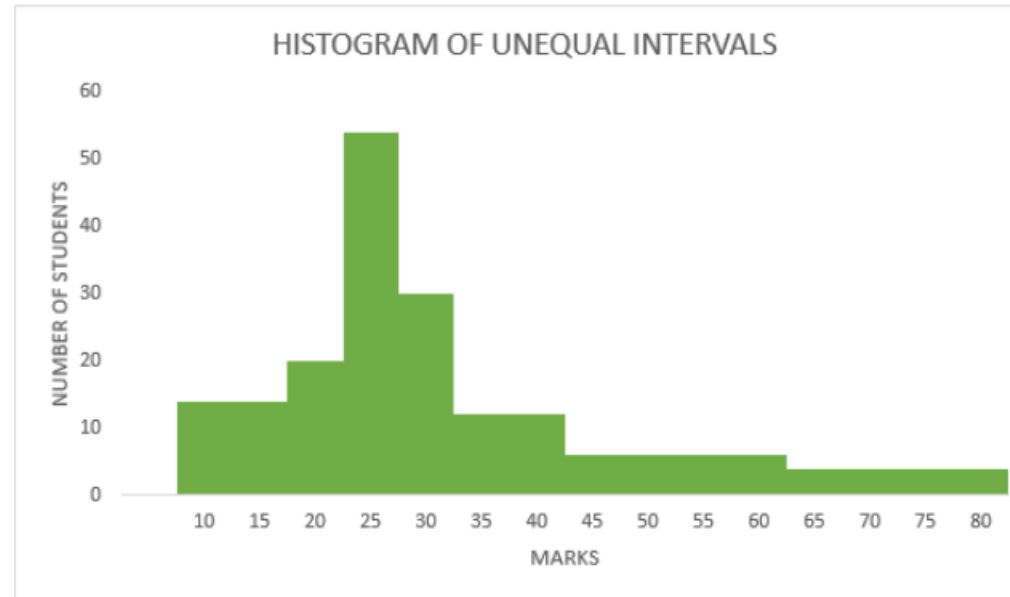


| Wages | Number of Workers | Adjustment Factor | Frequency Density<br>(Adjusted Frequency) |
|-------|-------------------|-------------------|-------------------------------------------|
| 10-15 | 14                | $5 \div 5 = 1$    | $14 \div 1 = 14$                          |
| 15-20 | 20                | $5 \div 5 = 1$    | $20 \div 1 = 20$                          |
| 20-25 | 54                | $5 \div 5 = 1$    | $54 \div 1 = 54$                          |
| 25-30 | 30                | $5 \div 5 = 1$    | $30 \div 1 = 30$                          |
| 30-40 | 24                | $10 \div 5 = 2$   | $24 \div 2 = 12$                          |
| 40-60 | 24                | $20 \div 5 = 4$   | $24 \div 4 = 6$                           |
| 60-80 | 16                | $20 \div 5 = 4$   | $16 \div 4 = 4$                           |

In the above table, the class interval is calculated as the difference between the upper-class limit and lower-class limit, i.e.,

$15-10=5$ ,  $20-15=5$ ,  $20-25=5$ ,  $30-25=5$ ,  $40-30=10$ ,  $60-40=20$ , and  $80-60=20$ .

#### 4. Plotting Histogram:



# Box Plot

- A Box Plot is also known as a Box and Whisker Plot and it is a graphical tool used to understand the distribution of numerical data. It shows the median, quartiles and possible outliers in a simple visual form. It helps compare distributions between groups with ease.

- multiple groups effectively.
- Identify outliers quickly and visually.
- Reveal data symmetry or skewness at a glance.
- Show variability clearly even for large datasets.
- Provide compact summaries of data using quartiles and range.
- Useful for comparing multiple groups or categories.

## Interquartile Range (IQR)

The box in a box plot represents the Interquartile Range (IQR).

$$\text{IQR} = Q_3 - Q_1$$

It covers the middle 50% of the data, making it a strong measure of spread that is resistant to extreme values.

## Identifying Outliers in a Box Plot

Outliers are values that lie outside the upper and lower limits:

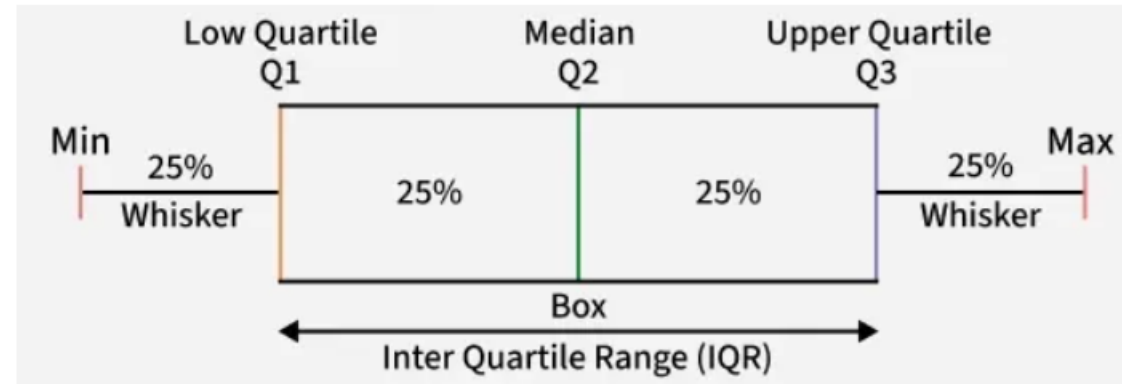
$$\text{Lower Limit} = Q_1 - 1.5 \times \text{IQR}$$

$$\text{Upper Limit} = Q_3 + 1.5 \times \text{IQR}$$

- Any value below the lower limit or above the upper limit is treated as an outlier and shown as a separate point.
- The minimum and maximum whiskers represent the closest values within these limits.

## Components of a Box Plot

A Box Plot visually represents the five-number summary of a dataset:



*Box Plot*

The box plot shown in the above diagram is a perfect plot with no skewness. The plots can have skewness and the median might not be at the center of the box.

- **Minimum:** The smallest non-outlier value shown at the lower whisker end.
- **First Quartile (Q1):** The value below which 25% of the data lies.
- **Median (Q2):** The middle value that splits the data into two equal halves.
- **Third Quartile (Q3):** The value below which 75% of the data lies.
- **Maximum:** The largest non-outlier value shown at the upper whisker end.

# How to Create a Box Plot

- Consider the runs scored by a cricket team in 12 matches: 100, 120, 110, 150, 110, 140, 130, 170, 120, 220, 140, 110
- **Step 1: Arrange Data in Ascending Order**
- 100, 110, 110, 110, 120, 120, 130, 140, 140, 150, 170, 220



### Step 2: Find Quartiles and Median

- Median ( $Q_2$ ) =  $\frac{120+130}{2} = 125$
- $Q_1$  = Median of first 6 values =  $\frac{110+110}{2} = 110$
- $Q_3$  = Median of last 6 values =  $\frac{140+150}{2} = 145$

### Step 3: Compute IQR

- $IQR = 145 - 110 = 35$

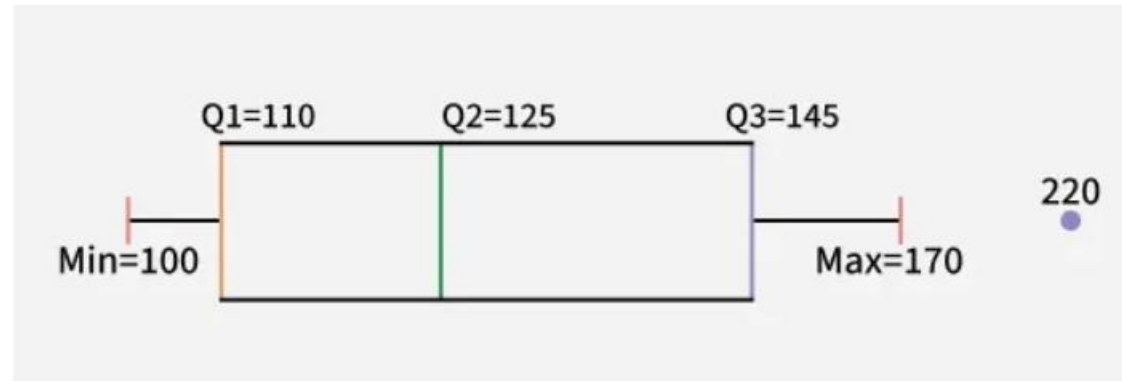
### Step 4: Calculate Limits

- Lower Limit =  $110 - 1.5 \times 35 = 57.5$
- Upper Limit =  $145 + 1.5 \times 35 = 197.5$

### Step 5: Identify Min, Max and Outliers

- Minimum (within limit) = 100
- Maximum (within limit) = 170
- Outlier = 220

This information is then used to draw the box plot.



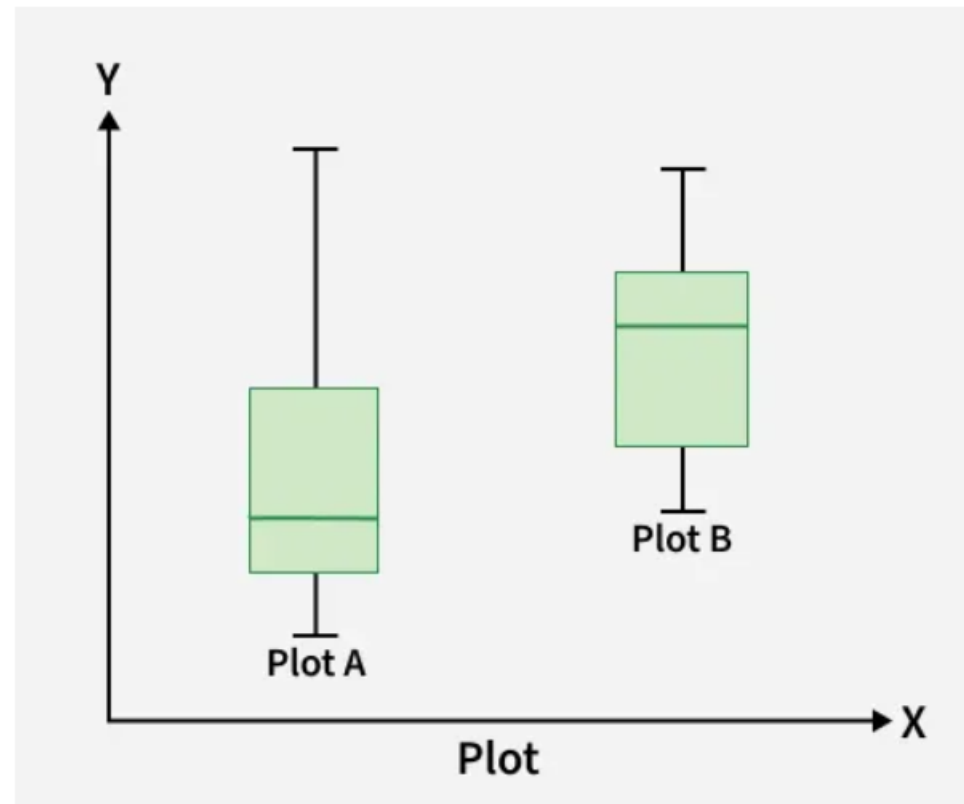
Box Plot

# Use-Cases of Box Plots

- Common use-cases include:
- **Exploratory Data Analysis (EDA):** Helps quickly detect spread, skewness and extreme values.
- **Comparing Multiple Groups:** Useful for visualizing differences in distributions across categories.
- **Quality Control and Process Monitoring:** Detects unusual deviations or anomalies in manufacturing and processes.
- **Statistical Reporting:** Provides a compact visual summary for large datasets.

## How to Compare Box Plots

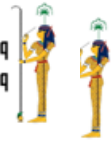
Comparing multiple box plots helps understand differences between groups in terms of center, spread, skewness and outliers.



Box Plot

- **1. Compare the Medians**
- Check if the median line of one plot lies outside the box of another.
- A median placed higher usually indicates larger overall values.
- If Group B's median falls outside Group A's box, the groups differ notably.

- **2. Compare the Dispersion (Spread)**
- IQR represents the height of the box, showing spread of middle 50 percent data.
- A longer box means higher variability.
- Longer whiskers indicate greater overall spread of values.
- Plot A with a bigger box and whiskers has more variable data.



- **3. Compare Outliers**
- Outliers appear as points beyond whiskers.
- More outliers indicate irregular or inconsistent data.
- Fewer outliers suggest more stable and predictable values.

- **4. Compare Skewness**

- Skewness shows asymmetry in data distribution.
- Median closer to the bottom with a longer upper whisker indicates right skew.
- Median near the top with a longer lower whisker indicates left skew.
- Plot A is left-skewed and Plot B is right-skewed, showing clear distribution differences.



## Difference Between Vertical and Horizontal Box Plot

Here we compare vertical and horizontal box plot

| Feature           | Vertical Box Plot                             | Horizontal Box Plot                                   |
|-------------------|-----------------------------------------------|-------------------------------------------------------|
| Orientation       | Box is drawn vertically along the y-axis      | Box is drawn horizontally along the x-axis            |
| Best For          | Comparing numerical values across categories. | Showing distribution when category labels are long.   |
| Readability       | Can become cramped with many categories.      | More readable when dataset has many labels.           |
| Common Use        | Default plot in most statistical tools.       | Useful for survey data, ratings and long text labels. |
| Space Utilization | Requires more vertical space.                 | Utilizes horizontal space more efficiently.           |

- **Advantages**
- **Easy Comparison:** Allows quick comparison of distributions across multiple groups.
- **Outlier Detection:** Clearly highlights unusual or extreme values in the data.
- **Quartile-Based Summary:** Summarizes data using median and quartiles instead of raw values.
- **Resistant to Noise:** Less affected by extreme values compared to mean based visuals.
- **Compact Representation:** Presents a large amount of statistical information in a clean, compact format.

- **Limitation**
- **No Distribution Shape:** Does not reveal whether data is normal, uniform or multi-peaked.
- **Mean Not Visible:** Shows median and quartiles but hides the average value.
- **Cannot Detect Multi-Modality:** Fails to identify multiple peaks or clusters in the data.
- **Less Useful for Small Samples:** Provides limited insights when dataset size is very small.

# Quantile Quantile plots

- The quantile-quantile( q-q plot) plot is a graphical method for determining if a dataset follows a certain probability distribution or whether two samples of data came from the same population or not.
- Q-Q plots are particularly useful for assessing whether a dataset is normally distributed or if it follows some other known distribution. They are commonly used in statistics, data analysis, and quality control to check assumptions and identify departures from expected distributions.

- **Quantiles And Percentiles**

- Quantiles are points in a dataset that divide the data into intervals containing equal probabilities or proportions of the total distribution. They are often used to describe the spread or distribution of a dataset. The most common quantiles are:

- **Median (50th percentile)**: The median is the middle value of a dataset when it is ordered from smallest to largest. It divides the dataset into two equal halves.
- **Quartiles (25th, 50th, and 75th percentiles)**: Quartiles divide the dataset into four equal parts. The first quartile (Q1) is the value below which 25% of the data falls, the second quartile (Q2) is the median, and the third quartile (Q3) is the value below which 75% of the data falls.
- **Percentiles**: Percentiles are similar to quartiles but divide the dataset into 100 equal parts. For example, the 90th percentile is the value below which 90% of the data falls.

- **Note:**
- A q-q plot is a plot of the quantiles of the first data set against the quantiles of the second data set.
- For reference purposes, a 45% line is also plotted; **For** if the samples are from the same population then the points are along this line.

## Normal Distribution:

The normal distribution (aka Gaussian distribution Bell curve) is a continuous probability distribution representing distribution obtained from the randomly generated real values.

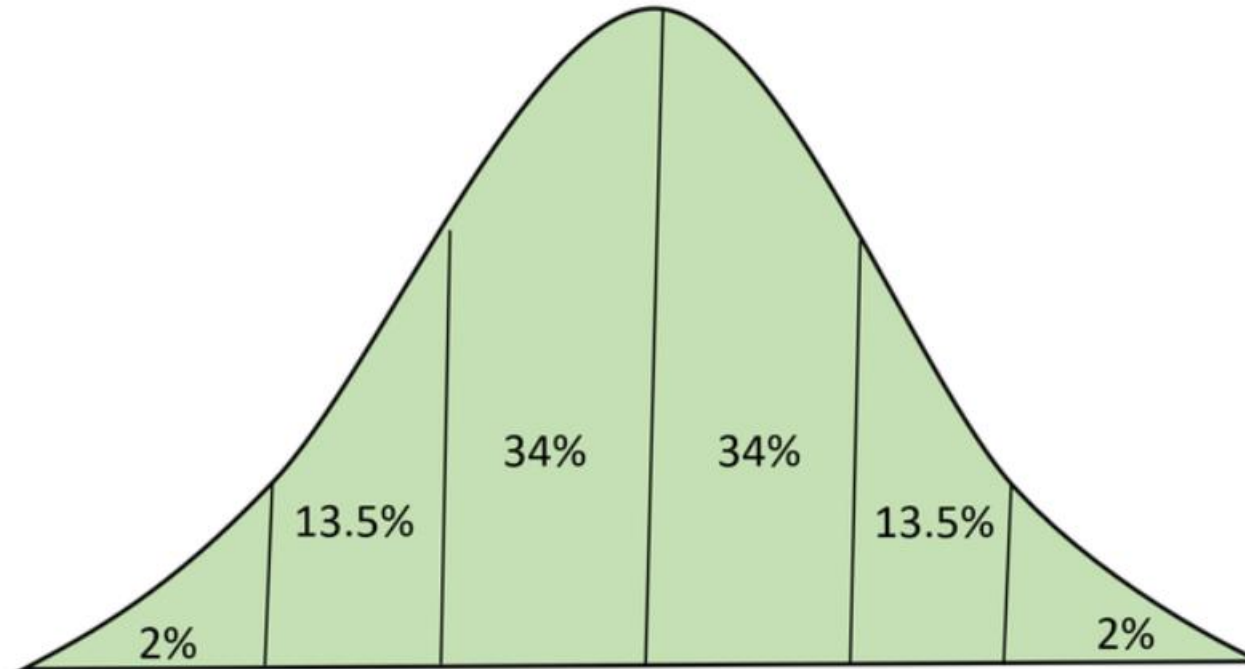
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$f(x)$  = *Probability Distribution Function*

$\mu$  = *mean*

$\sigma$  = *standard deviation*





*Normal Distribution with Area Under Curve*

- **How to Draw Q-Q plot?**
- To draw a Quantile-Quantile (Q-Q) plot, you can follow these steps:
- **Collect the Data:** Gather the dataset for which you want to create the Q-Q plot. Ensure that the data are numerical and represent a random sample from the population of interest.

- **Sort the Data:** Arrange the data in either ascending or descending order. This step is essential for computing quantiles accurately.
- **Choose a Theoretical Distribution:** Determine the theoretical distribution against which you want to compare your dataset. Common choices include the normal distribution, exponential distribution, or any other distribution that fits your data well.
- **Calculate Theoretical Quantiles:** Compute the quantiles for the chosen theoretical distribution. For example, if you're comparing against a normal distribution, you would use the inverse cumulative distribution function (CDF) of the normal distribution to find the expected quantiles.

## • Plotting:

- Plot the sorted dataset values on the x-axis.
- Plot the corresponding theoretical quantiles on the y-axis.
- Each data point (x, y) represents a pair of observed and expected values.
- Connect the data points to visually inspect the relationship between the dataset and the theoretical distribution.

- **Interpretation of Q-Q plot**
- If the points on the plot fall approximately along a straight line, it suggests that your dataset follows the assumed distribution.
- Deviations from the straight line indicate departures from the assumed distribution, requiring further investigation.

- **Advantages of Q-Q plot**
- **Flexible Comparison:** Q-Q plots can compare datasets of different sizes without **requiring equal sample sizes**.
- **Dimensionless Analysis:** They are dimensionless, making them suitable for comparing datasets with **different units or scales**.

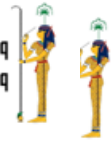
- **Visual Interpretation:** Provides a clear visual representation of data distribution compared to a theoretical distribution.
- **Sensitive to Deviations:** Easily detects departures from assumed distributions, aiding in identifying data discrepancies.
- **Diagnostic Tool:** Helps in assessing distributional assumptions, identifying outliers, and understanding data patterns.

- **Applications Of Quantile-Quantile Plot**
- The Quantile-Quantile plot is used for the following purpose:
- **Assessing Distributional Assumptions:** Q-Q plots are frequently used to visually inspect whether a dataset follows a specific probability distribution, such as the normal distribution. By comparing the quantiles of the observed data to the quantiles of the assumed distribution, deviations from the assumed distribution can be detected. This is crucial in many statistical analyses, where the validity of distributional assumptions impacts the accuracy of statistical inferences.



- **Detecting Outliers:** Outliers are data points that deviate significantly from the rest of the dataset. Q-Q plots can help identify outliers by revealing data points that fall far from the expected pattern of the distribution. Outliers may appear as points that deviate from the expected straight line in the plot.
- **Comparing Distributions:** Q-Q plots can be used to compare two datasets to see if they come from the same distribution. This is achieved by plotting the quantiles of one dataset against the quantiles of another dataset. If the points fall approximately along a straight line, it suggests that the two datasets are drawn from the same distribution.

- **Assessing Normality:** Q-Q plots are particularly useful for assessing the normality of a dataset. If the data points in the plot closely follow a straight line, it indicates that the dataset is approximately normally distributed. Deviations from the line suggest departures from normality, which may require further investigation or non-parametric statistical techniques.
- **Model Validation:** In fields like econometrics and machine learning, Q-Q plots are used to validate predictive models. By comparing the quantiles of observed responses with the quantiles predicted by a model, one can assess how well the model fits the data. Deviations from the expected pattern may indicate areas where the model needs improvement.
- **Quality Control:** Q-Q plots are employed in quality control processes to monitor the distribution of measured or observed values over time or across different batches. Departures from expected patterns in the plot may signal changes in the underlying processes, prompting further investigation.



- Types of Q-Q plots
- There are several types of Q-Q plots commonly used in statistics and data analysis, each suited to different scenarios or purposes:

- Normal Distribution: A symmetric distribution where the Q-Q plot would show points approximately along a diagonal line if the data adheres to a normal distribution.
- Right-skewed Distribution: A distribution where the Q-Q plot would display a pattern where the observed quantiles deviate from the straight line towards the upper end, indicating a longer tail on the right side.

- Left-skewed Distribution: A distribution where the Q-Q plot would exhibit a pattern where the observed quantiles deviate from the straight line towards the lower end, indicating a longer tail on the left side.
- Under-dispersed Distribution: A distribution where the Q-Q plot would show observed quantiles clustered more tightly around the diagonal line compared to the theoretical quantiles, suggesting lower variance.

- Over-dispersed Distribution: A distribution where the Q-Q plot would display observed quantiles more spread out or deviating from the diagonal line, indicating higher variance or dispersion compared to the theoretical distribution.



# Thank You